Newman-Penrose Formalism Calculations

Stephani et al., Equation (28.56a)

Metric

$$ds^2 = \left(-\frac{\kappa_0 e^x q^2 - 4\,mr}{2\,r^2}\right) \mathrm{d}u \otimes \mathrm{d}u - \mathrm{d}u \otimes \mathrm{d}r - \mathrm{d}r \otimes \mathrm{d}u + \frac{r^2}{P^2} \mathrm{d}x \otimes \mathrm{d}x + \frac{r^2}{P^2} \mathrm{d}y \otimes \mathrm{d}y$$

Coordinates

$$(x^\mu) = (u, r, x, y)$$

Metric Functions

$$P(x) = P$$

$$q(u) = q$$

Null Tetrad

Covariant components

$$l = (1)\ du$$

$$n = \left(\frac{\kappa_0 e^x q^2}{4\,r^2} - \frac{m}{r}\right)\ du + (1)\ dr$$

$$m = \left(\frac{\sqrt{2}r}{2\,P}\right)\ dx + \left(-\frac{i\,\sqrt{2}r}{2\,P}\right)\ dy$$

$$\overline{m} = \left(\frac{\sqrt{2}r}{2\,P}\right)\ dx + \left(\frac{i\,\sqrt{2}r}{2\,P}\right)\ dy$$

Contravariant components

$$l = (-1)\ dr$$

$$n = (-1)\ du + \left(\frac{\kappa_0 e^x q^2}{4\,r^2} - \frac{m}{r}\right)\ dr$$

$$m = \left(\frac{\sqrt{2}P}{2\,r}\right)\ dx + \left(-\frac{i\,\sqrt{2}P}{2\,r}\right)\ dy$$

$$\overline{m} = \left(\frac{\sqrt{2}P}{2\,r}\right)\ dx + \left(\frac{i\,\sqrt{2}P}{2\,r}\right)\ dy$$

Directional Derivatives

$$D = -\partial_r$$

$$\Delta = -\partial_u + \frac{\kappa_0 e^x q^2}{4\,r^2} - \frac{m}{r}\,\partial_r$$

$$\delta = \frac{\sqrt{2}P}{2\,r}\,\partial_x - \frac{i\,\sqrt{2}P}{2\,r}\,\partial_y$$

$$\overline{\delta} = \frac{\sqrt{2}P}{2\,r}\,\partial_x + \frac{i\,\sqrt{2}P}{2\,r}\,\partial_y$$

Spin Coefficients

$$\rho = \frac{1}{r}$$

$$\sigma = 0$$

$$\kappa = 0$$

$$\mu = \frac{\kappa_0 e^x q^2}{4\, r^3} - \frac{m}{r^2}$$

$$\lambda = 0$$

$$\nu = \frac{\sqrt{2}\kappa_0 P e^x q^2}{8\, r^3}$$

$$\alpha = \frac{\sqrt{2}\frac{\partial}{\partial x} P}{4\, r}$$

$$\beta = -\frac{\sqrt{2}\frac{\partial}{\partial x} P}{4\, r}$$

$$\tau = 0$$

$$\pi = 0$$

$$\epsilon = 0$$

$$\gamma = \frac{\kappa_0 e^x q^2}{4\, r^3} - \frac{m}{2\, r^2}$$

Ricci Tensor Components

$$\Phi_{00} = 0$$

$$\Phi_{01} = 0$$

$$\Phi_{02} = 0$$

$$\Phi_{10} = 0$$

$$\Phi_{11} = \frac{\kappa_0 e^x q^2}{4\,r^4} - \frac{\frac{\partial}{\partial x} P^2}{4\,r^2} + \frac{P \frac{\partial^2}{(\partial x)^2} P}{4\,r^2}$$

$$\Phi_{12} = \frac{\sqrt{2}\kappa_0 P e^x q^2}{8\,r^4}$$

$$\Phi_{20} = 0$$

$$\Phi_{21} = \frac{\sqrt{2}\kappa_0 P e^x q^2}{8\,r^4}$$

$$\Phi_{22} = \frac{\kappa_0 P^2 e^x q^2}{8\,r^4} + \frac{\kappa_0 e^x q \frac{\partial}{\partial u} q}{2\,r^3}$$

$$\Lambda = -\frac{\frac{\partial}{\partial x} P^2}{12\,r^2} + \frac{P \frac{\partial^2}{(\partial x)^2} P}{12\,r^2}$$

Weyl Tensor Components

$$\Psi_0 = 0$$

$$\Psi_1 = 0$$

$$\Psi_2 = \frac{\kappa_0 e^x q^2}{2\, r^4} + \frac{\frac{\partial}{\partial x} P^2}{6\, r^2} - \frac{P \frac{\partial^2}{(\partial x)^2} P}{6\, r^2} - \frac{m}{r^3}$$

$$\Psi_3 = \frac{\sqrt{2}\kappa_0 P e^x q^2}{4\, r^4}$$

$$\Psi_4 = \frac{\kappa_0 P^2 e^x q^2}{8\, r^4} + \frac{\kappa_0 P e^x q^2 \frac{\partial}{\partial x} P}{4\, r^4}$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "II"